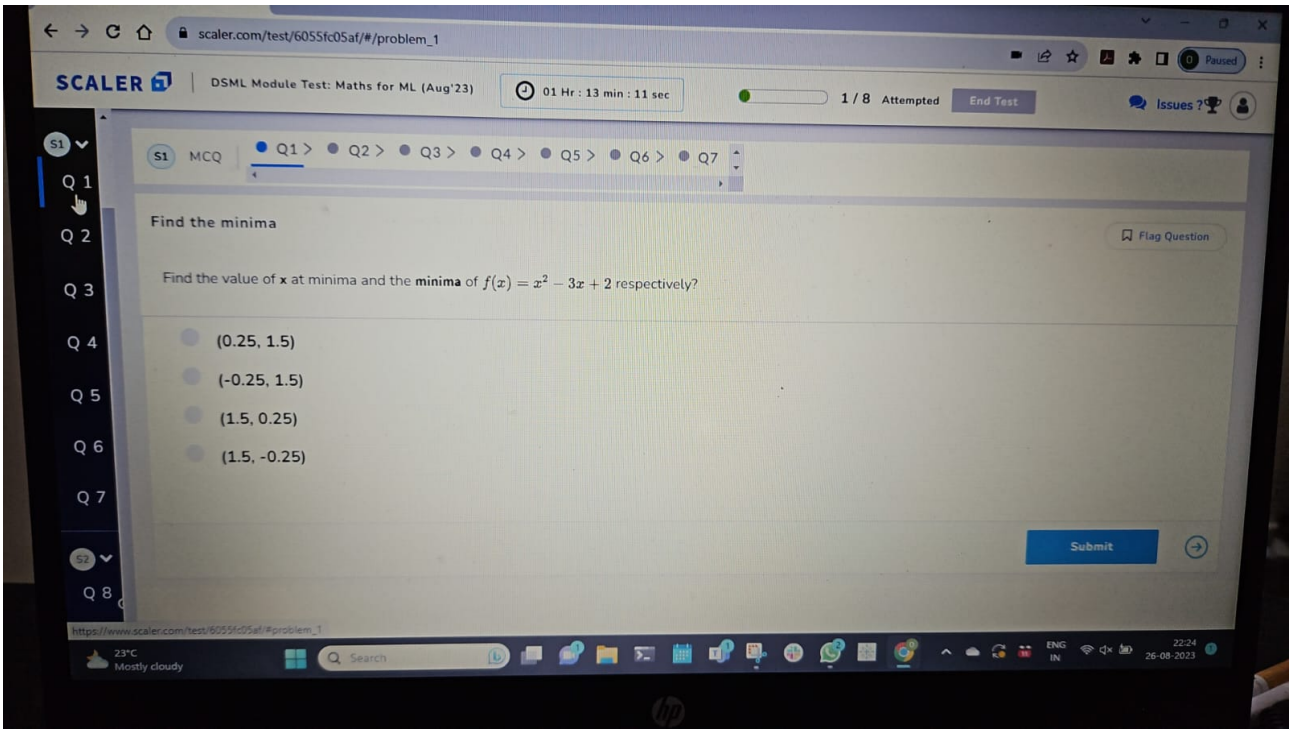
Ratnesh MATHS FOR ML (18 AUG to 27 AUG 23)

DSML MODULE TEST

MATHS FOR ML (18 AUG to 27 AUG 23)

MATHS FOR Machine Learning

Q1. FIND THE MINIMA (1.5 , -0.25)



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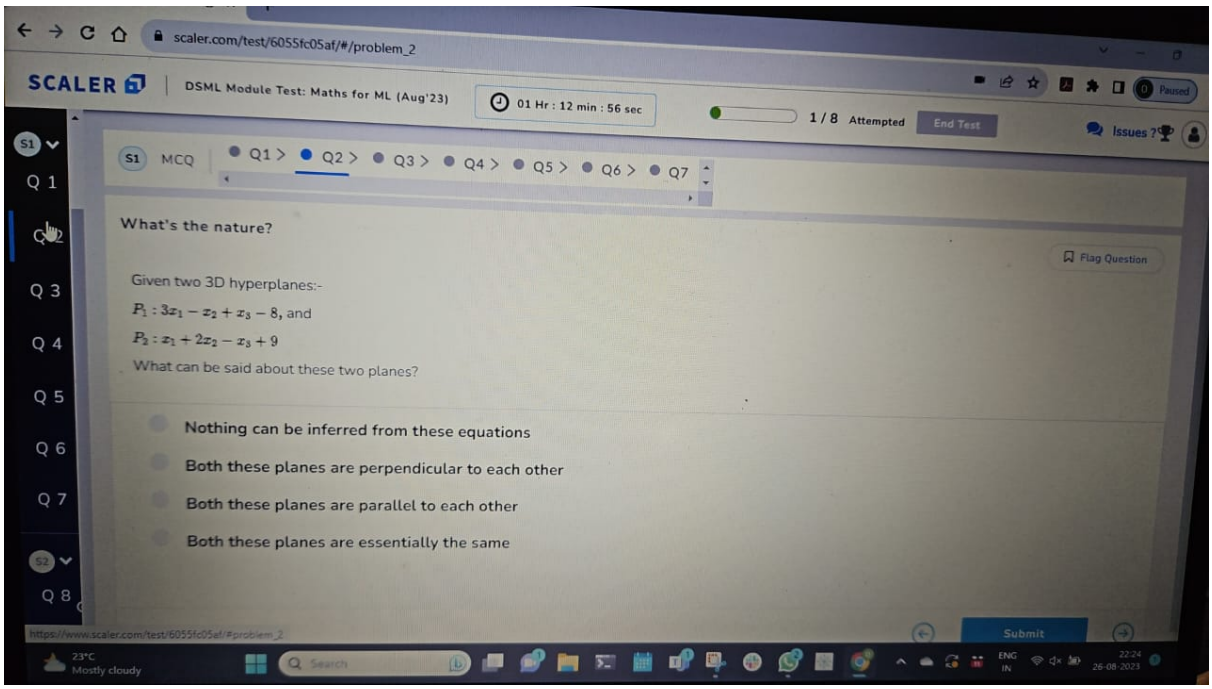
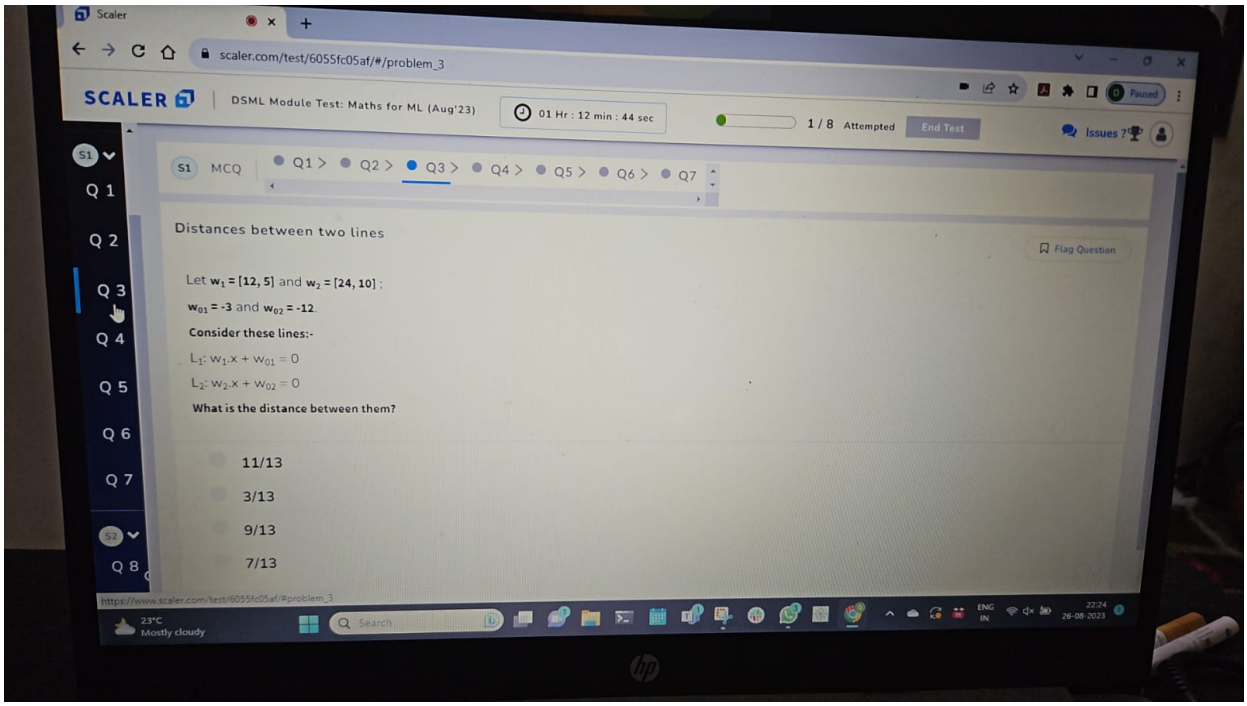
Q2. WHAT'S THE NATURE (both these are perpendicular to each other)



Q3. DISTANCE BETWEEN TWO LINES (3/13)



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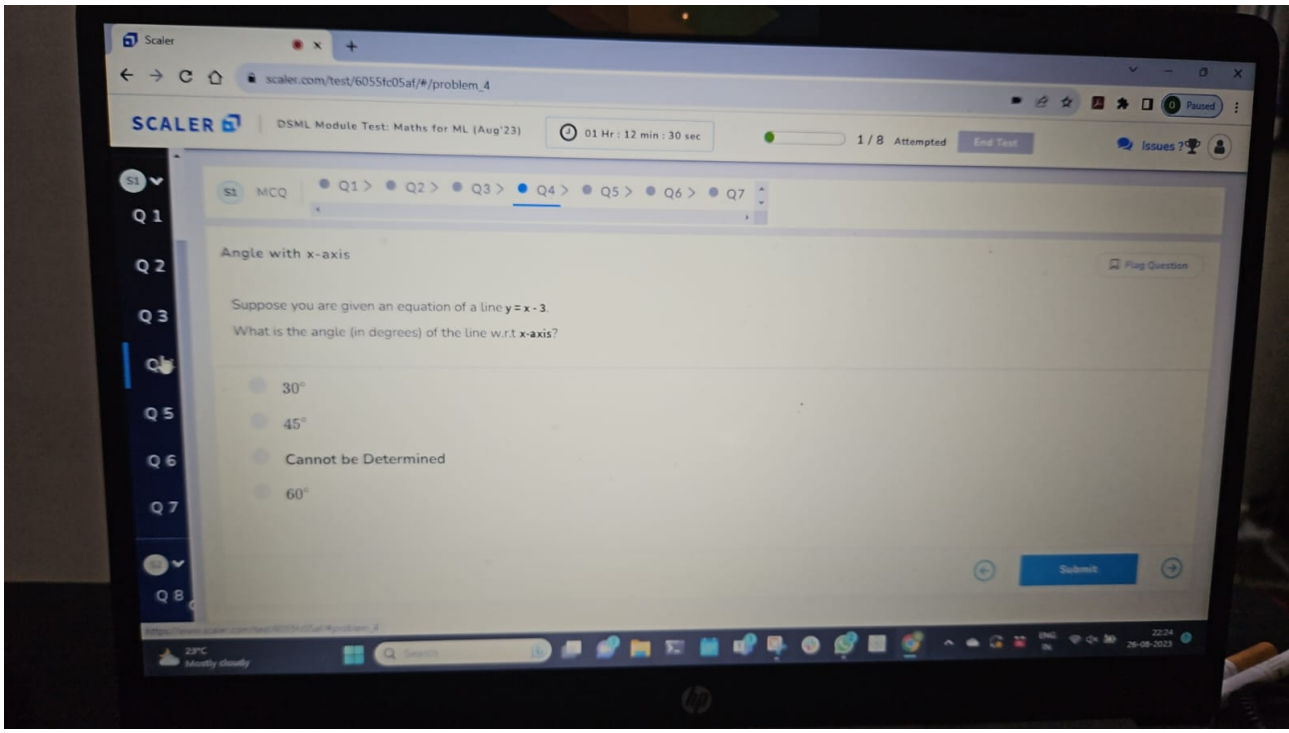
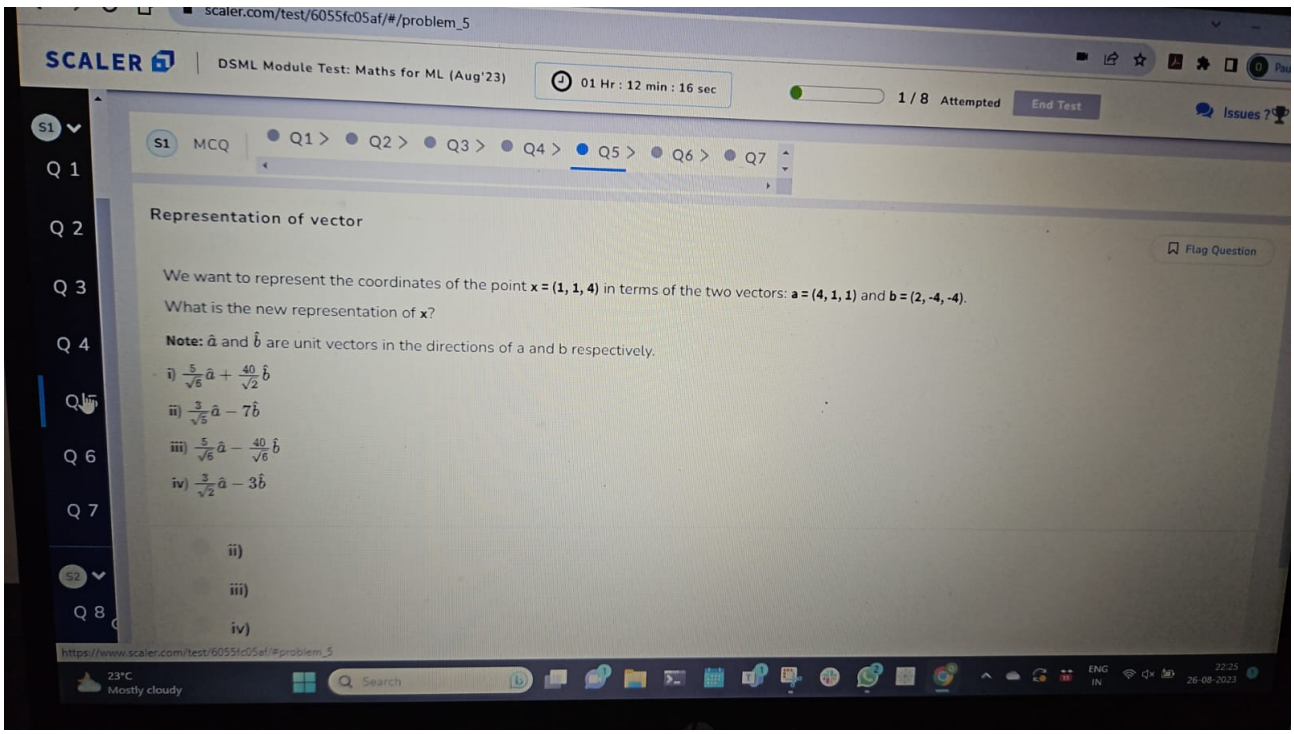
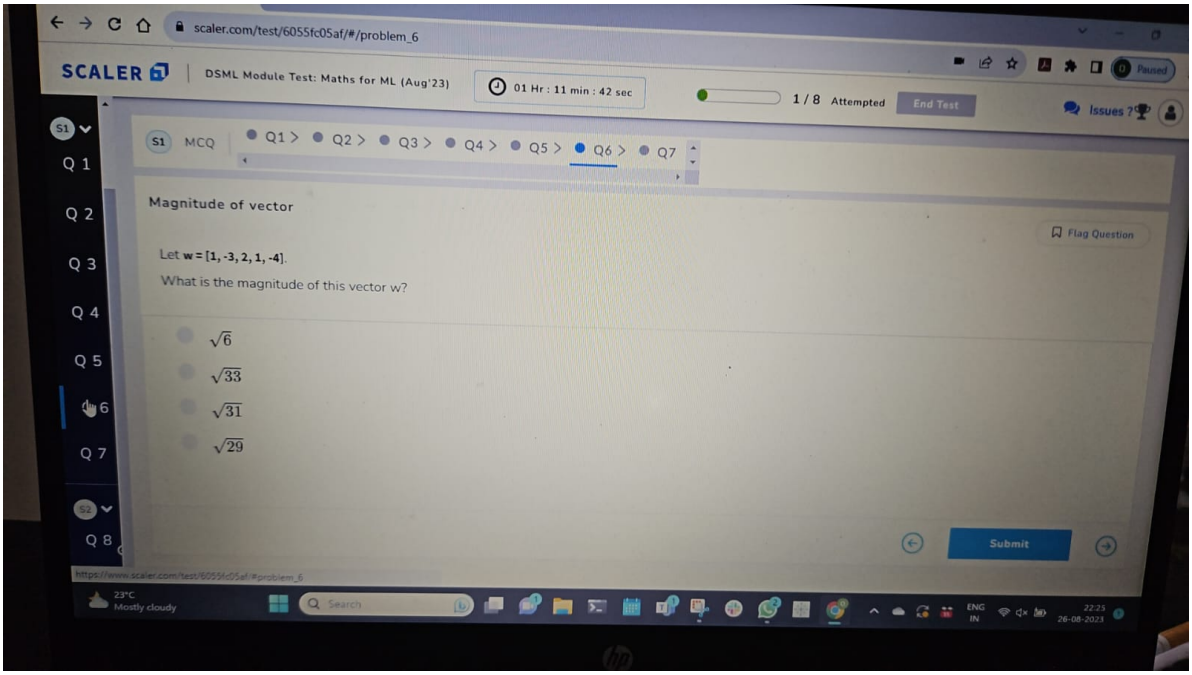
Q4. ANGLE WITH X-AXIS (45 degree)



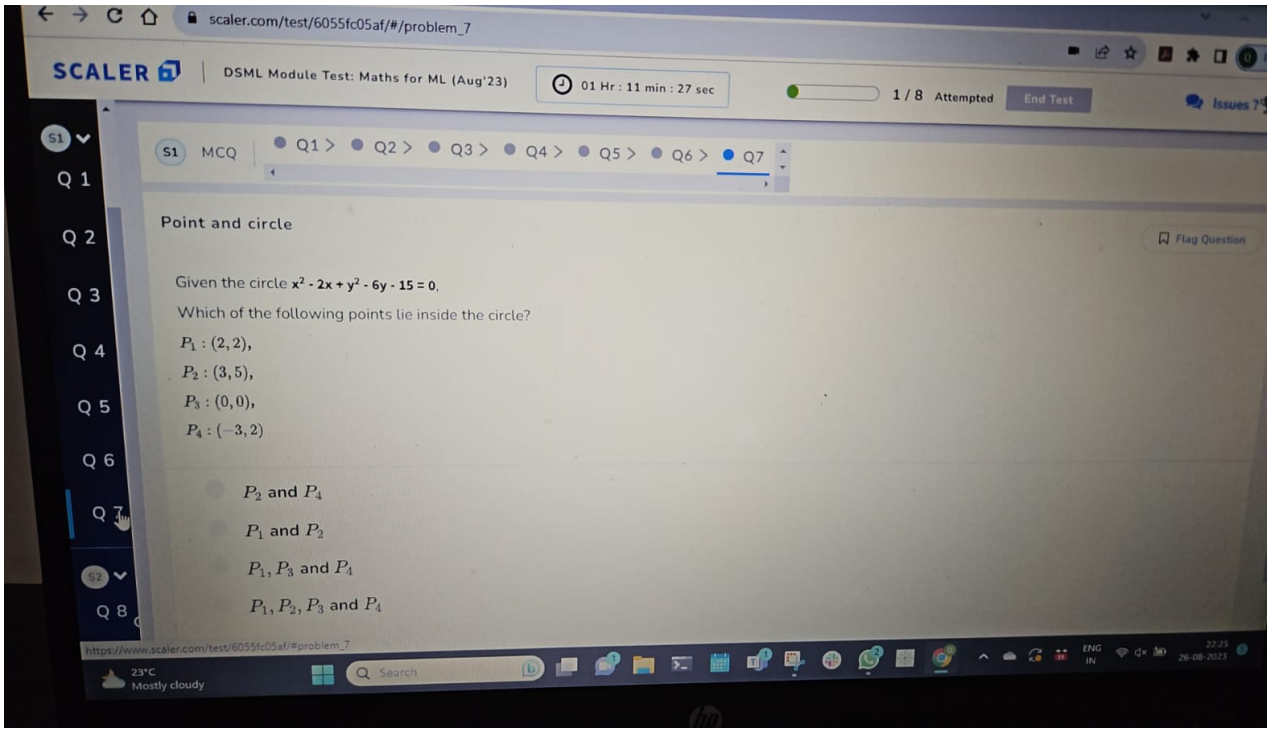
Q5. REPRESENTATION OF VECTOR (iv option)



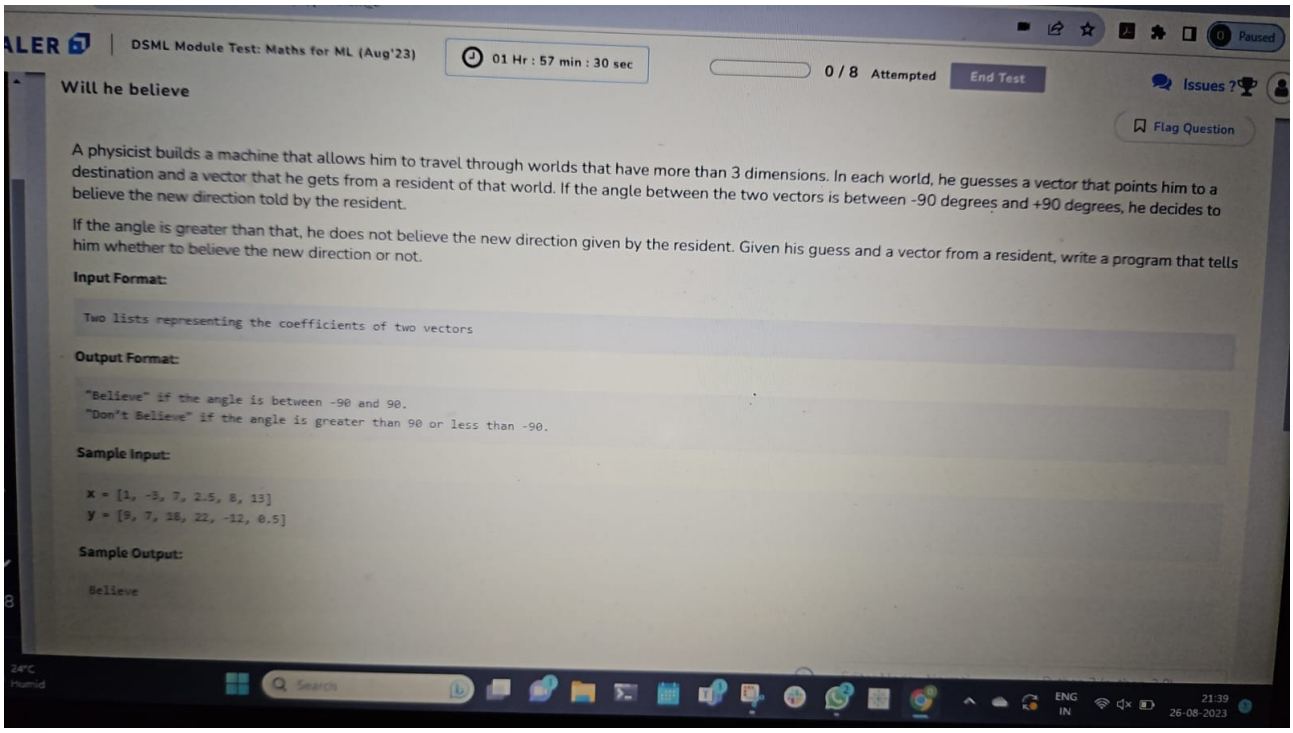
Q6. MAGNITUDE OF VECTOR sqrt(31)



Q7. POINT and CIRLCE (P1, P2, P3, P4)



Q8. WILL HE BELIEVE



DSML Module Test: Maths for ML (Aug'23)

01 Hr : 57 min : 30 sec

0 / 8 Attempted

End Test

Issues ?

Flag Question

Will he believe

A physicist builds a machine that allows him to travel through worlds that have more than 3 dimensions. In each world, he guesses a vector that points him to a destination and a vector that he gets from a resident of that world. If the angle between the two vectors is between -90 degrees and +90 degrees, he decides to believe the new direction told by the resident.

If the angle is greater than that, he does not believe the new direction given by the resident. Given his guess and a vector from a resident, write a program that tells him whether to believe the new direction or not.

Input Format:

```
Two lists representing the coefficients of two vectors
```

Output Format:

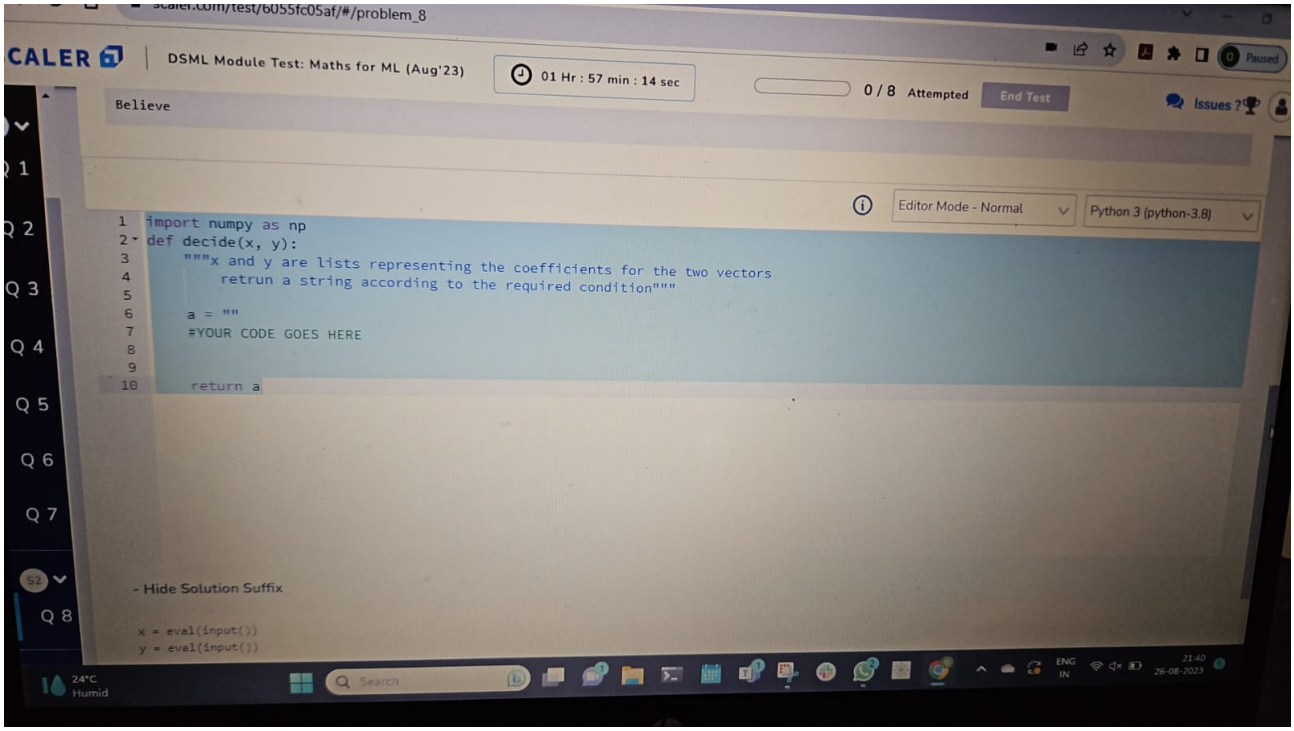
```
"Believe" if the angle is between -90 and 90.
"Don't Believe" if the angle is greater than 90 or less than -90.
```

Sample Input:

```
x = [1, -3, 7, 2.5, 8, 13]
y = [9, 7, 18, 22, -12, 0.5]
```

Sample Output:

```
Believe
```



DSML Module Test: Maths for ML (Aug'23)

01 Hr : 57 min : 14 sec

0 / 8 Attempted

End Test

Issues ?

Believe

Editor Mode - Normal

Python 3 (python-3.8)

```
import numpy as np
def decide(x, y):
    """x and y are lists representing the coefficients for the two vectors
        retrun a string according to the required condition"""

    a = ""
    #YOUR CODE GOES HERE


    return a
```

- Hide Solution Suffix

```
x = eval(input())
y = eval(input())
```

Will he believe

A physicist builds a machine that allows him to travel through worlds that have more than 3 dimensions. In each world, he guesses a vector that points him to a destination and a vector that he gets from a resident of that world. If the angle between the two vectors is between -90 degrees and +90 degrees, he decides to believe the new direction told by the resident.

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Input Format:

Tull lists representing the coefficients of two vectors

Output Format:

"Believe" if the angle is between -90 and 90. "Don't Believe" if the angle is greater than 90 or less than -90.

Sample Input:
x= [1, -3, 7, 2.5, 8, 13]
y=[9, 7, 18, 22, -12, 0.5]
Sample Output:
Believe

```
import math
import numpy as np

def decide(x, y):

  dot_product = sum(x[i] * y[i] for i in range(len(x)))
  magnitude_x = math.sqrt(sum(x[i] ** 2 for i in range(len(x))))
  magnitude_y = math.sqrt(sum(y[i] ** 2 for i in range(len(y))))
  cos = (dot_product / (magnitude_x * magnitude_y))

  if 0<= cos<=1:
    return "Believe"
  else:
    return "Don't Believe “
```

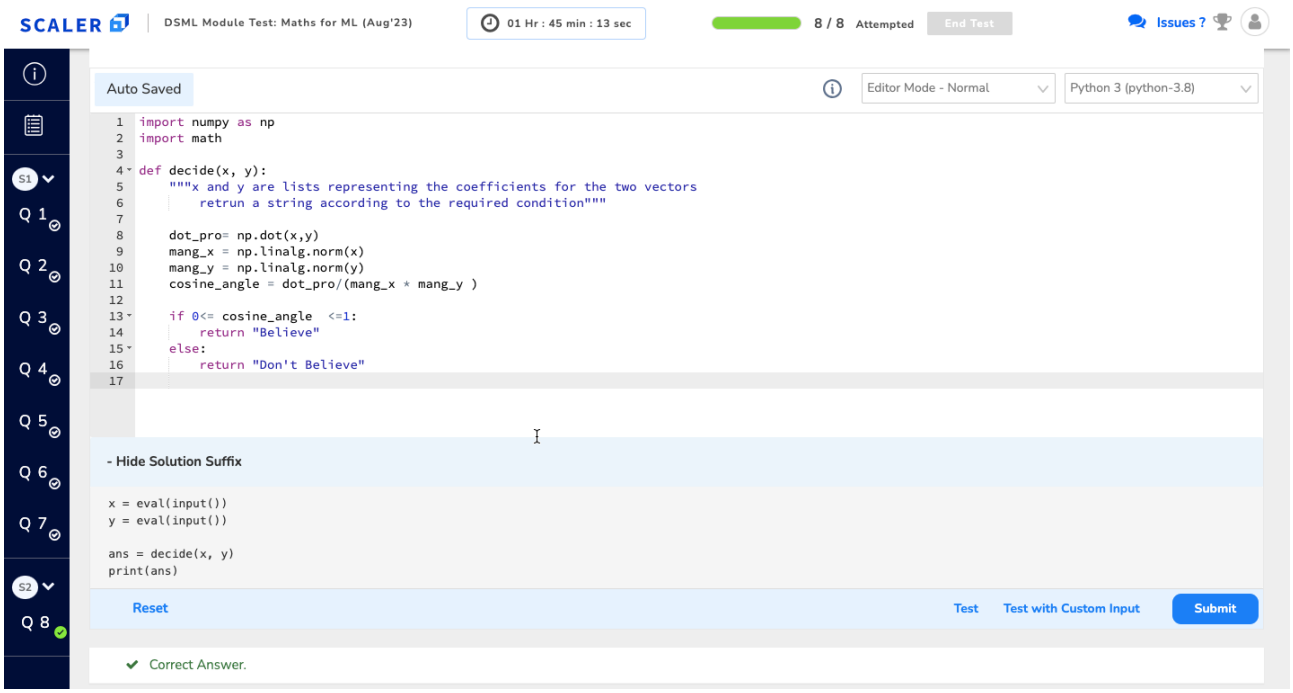
Or

```
import numpy as np
def decide(x, y):
    x = np.array(x)
    y = np.array(y)

    dot_product = np.dot(x, y)

    magnitude_x = np.linalg.norm(x)
    magnitude_y = np.linalg.norm(y)

    cosine_angle = dot_product / (magnitude_x * magnitude_y)

    if 0 <=  cosine_angle  <= 1:
        return "Believe"
    else:
        return "Don't Believe"
```

Sol :